

# IBRAHIM ROGERS

MECHANICAL ENGINEERING | TEST SYSTEMS & METROLOGY | HARDWARE DEVELOPMENT

B.S. Mechanical Engineering, Northeastern University · Expected May 2028  
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## SELECTED WORK

|    |                                                   |                        |
|----|---------------------------------------------------|------------------------|
| 01 | Liquid Valve Endurance Tester & Leak-Test Fixture | Parker Hannifin · 2026 |
| 02 | Contamination Root-Cause Investigations           | Parker Hannifin · 2026 |
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# Liquid Valve Endurance Tester & Leak-Test Fixture

Quality & Sustaining Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

## OVERVIEW

**Context:** Precision fluid-control valves for medical and life-science systems. My work spans test-system design, fixture design, and root-cause investigation on returned product.

**My Role:** Requirements definition and hydraulic architecture for a closed-loop endurance tester; mechanical design of a supplier-side leak-test fixture.

## CLOSED-LOOP DI-WATER VALVE ENDURANCE TESTER

Consolidated conflicting inputs from quality, manufacturing, and engineering into a single defined test-system scope, then developed the architecture and produced build-ready drawings, schematics, and P&IDs to match.

|                            |                                    |
|----------------------------|------------------------------------|
| Test capacity              | 30–60 valves cycled simultaneously |
| Flow per valve / aggregate | 0.5 L/min / 15–30 L/min            |
| Operating pressure         | 120 psig                           |
| Fluid temperature control  | ±5°F of ambient                    |
| Working fluid              | DI water, continuous recirculation |

Sized pump, chiller, reservoir, manifold, and instrumentation, and integrated pressure control, relief protection, filtration, and a chiller bypass loop for thermal management under continuous recirculation. The loop is fluid-agnostic downstream of the reservoir, so the test rack does not need redesign if the working fluid changes.

## PRESSURE-DECAY LEAK-TEST FIXTURE

Designed a fixture in Autodesk Inventor for supplier-side incoming inspection of a molded fluid-control component.

|                    |                                            |
|--------------------|--------------------------------------------|
| Leak resolution    | Below 0.1 sccm at the orifice              |
| Trapped volume     | Under 0.5 cc                               |
| Method selected    | Pressure decay                             |
| Seal architecture  | Guarded three-port                         |
| Probe sizing basis | Worst-case tolerance (±0.001 in) + MMC/LMC |

## DESIGN REASONING

I selected pressure decay over steady-state make-up flow because commercially available flow instrumentation could not resolve the sub-0.1 sccm requirement at acceptable cost. Locating and sealing functions are separated so the probe seals against the orifice without over-constraining a molded part that carries real dimensional variation.

# Contamination Root-Cause Investigations

Quality & Sustaining Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

## OVERVIEW

Led two independent root-cause investigations into recurring return failures on precision fluid-control valves. Each involved a distinct contamination mechanism and a separate population of returned parts.

### INVESTIGATION A — ORIFICE CORROSION

|                    |                                                     |
|--------------------|-----------------------------------------------------|
| Scope              | 500+ returned valve bodies                          |
| Variables isolated | Supplier, stock, cleaning, floor process            |
| Outcome            | Contamination source isolated; mitigation validated |

I designed isolation experiments across each candidate variable instead of inspecting finished returns alone. That let the investigation separate a supplier-side cause from a downstream cleaning or floor-process cause.

### INVESTIGATION B — FOREIGN-OBJECT CONTAMINATION

|        |                                                              |
|--------|--------------------------------------------------------------|
| Scope  | 100+ returned units                                          |
| Method | Teardown and comparative inspection against conforming parts |

A separate contamination mechanism from Investigation A. It required its own teardown protocol and comparison baseline; the corrosion investigation's isolation variables did not transfer.

#### WHAT I TOOK FROM IT

- **Two investigations, not one.** Treating these as a single "contamination" bucket would have averaged two different mechanisms into one wrong answer. Splitting them is what let each get isolated correctly.
- **Comparative inspection needs a real baseline.** Conforming parts had to be inspected alongside the returns to make "different" mean anything.
- **Mitigation validation is a process.** Confirming a fix takes sustained data, not a single passing result.

# Packaging Redesign & Machine-Vision Inspection

Quality & Sustaining Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

## PACKAGING REDESIGN — TRANSIT DAMAGE ON A LOW-VOLUME COMPONENT

|              |                                          |
|--------------|------------------------------------------|
| Trigger      | Field-reported transit damage            |
| Part profile | Low-volume, high-value                   |
| Constraint   | Fixed cost ceiling set by unit economics |

The binding constraint here was economic, not technical. A packaging fix that eliminated damage but cost more than the value it protected would never have shipped, so the design had to converge on something effective enough to stop the transit failure mode while staying inside the cost ceiling.

## MACHINE-VISION INSPECTION SYSTEM — IN DEVELOPMENT

|        |                                                   |
|--------|---------------------------------------------------|
| Goal   | Flag defects on a production cell before shipment |
| Stack  | Python, OpenCV                                    |
| Status | In development                                    |

This extends the computer-vision work from my pan/tilt tracking system (Project 05) into a production application. A production inspection system has to run unattended against real part variation, which is a harder detection problem than tracking a fiducial marker under controlled lighting.

# Refreshable Braille Tactile Display

Hardware Product Lab — Forge, Northeastern University · Jan 2026 – Jun 2026

## OVERVIEW

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**Challenge:** Build a refreshable Braille cell that is low-cost, survives thousands of actuation cycles, and is small enough to tile into a line of text. The governing constraint is dimensional: a six-dot cell spans 6.25 mm, and any motor worth using is wider than the cell it drives.

**My Role:** Mechanism architecture and encoding scheme, CAD, full drawing package, clearance and tolerance-stack iteration, prototype fabrication, and cycle testing.

## DESIGN PROGRESSION

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The cell went through three drive architectures. Each change was driven by a constraint the previous one could not meet.

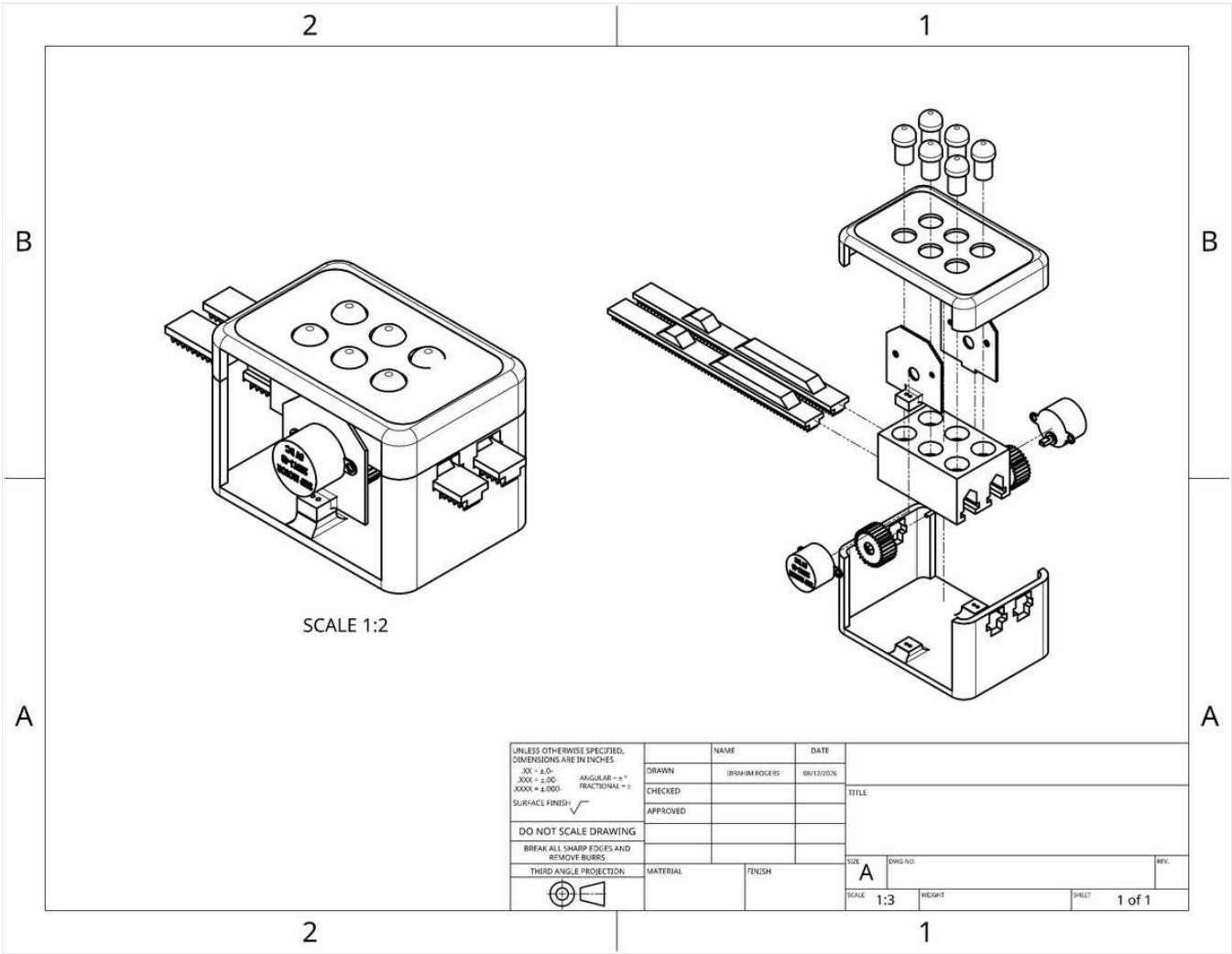
|                          |                                                                                                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 — Lead screw           | A lead screw integrated directly into the key system, chosen to minimize packaging by putting the drive inside the cell envelope. Dropped on time and cost before it was built. |
| 2 — Direct rack & pinion | One motor and pinion driving a single rack beneath the pin block, lifting six dome pins off cam profiles. This is the build that was drawn, fabricated, and cycle-tested.       |
| 3 — Bevel geartrain      | Two racks per cell driven through a bevel geartrain from outboard motors, with each rack encoding all eight states of its three pins. The architecture the project ended on.    |

The pivot from the lead screw was a schedule and budget decision rather than a technical one, and it cost packaging density. The move from direct rack and pinion to the bevel geartrain was the opposite: it was made specifically to solve tiling, which the direct-drive layout could not do at reading pitch.

The pages that follow run in that order — the direct-drive build and its drawing package first, then the bevel geartrain architecture and the at-size hardware.

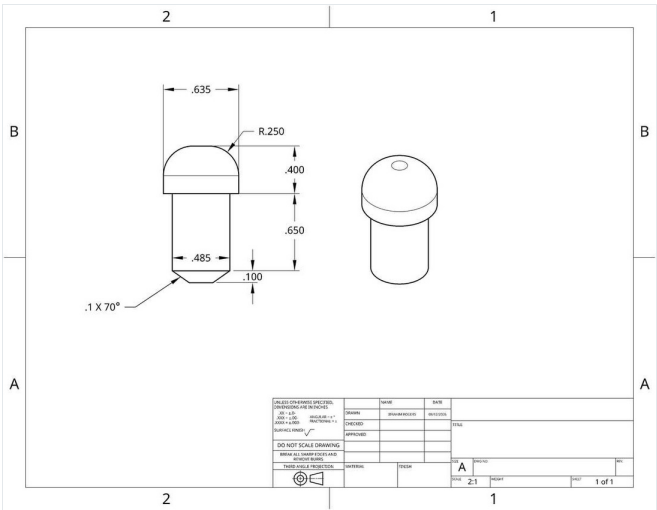
After the lead-screw concept was dropped, the cell was built as a direct-drive mechanism at enlarged scale: one motor and pinion advancing a single rack beneath the pin block, with each dot carried on its own dome pin. This is the version that was released as a drawing package, fabricated, and cycled. The drawings, tolerance work, and failure analysis on the next three pages all come from it.

FULL ASSEMBLY — EXPLODED & ISOMETRIC

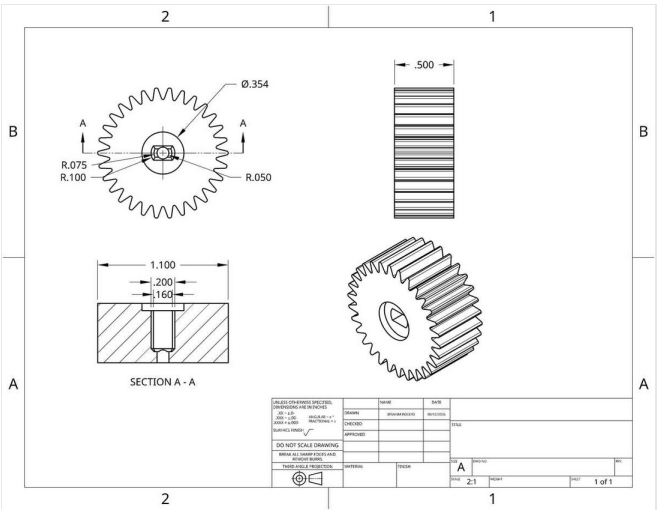


Top-level assembly drawing. Left: assembled enclosure (Scale 1:2). Right: exploded view (Scale 1:3) showing the six dome pins, letter-housing insert, guide bracket, pin-guide block, drive motor and pinion, rack, and enclosure top/bottom. Self-authored release drawing, dimensioned in inches per standard third-angle projection.

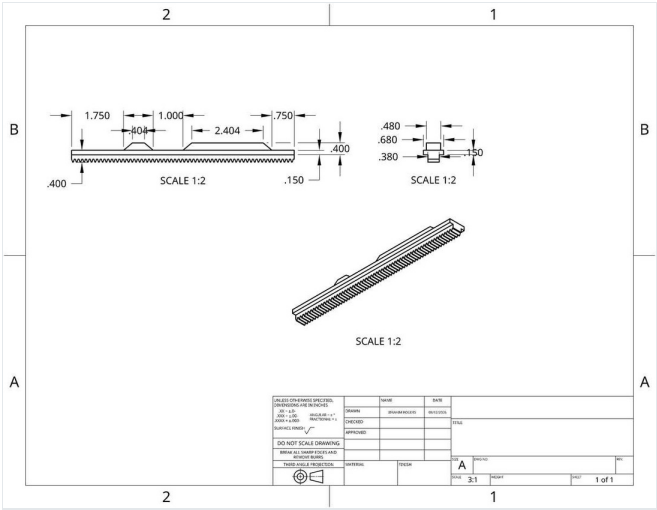
ACTUATION TRAIN — PIN, GEAR, RACK, DOWEL



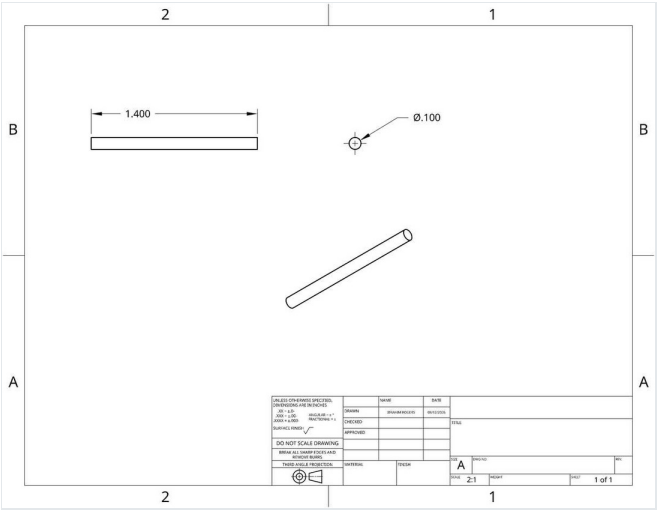
**Dome pin.** Ø.485 in shaft with a Ø.635 in domed head (R.250 radius) and a .1 × 70° lead-in chamfer at the tip. The shaft rides in the Ø.500 in bore of the pin-guide block, giving .015 in diametral clearance across the actuation stroke.



**Drive pinion.** Ø1.100 in spur gear with a Ø.354 in bore and a keyed drive slot (.200/.160 in) for the motor shaft. Root and fillet radii (R.050–R.100) are sized for the FDM-printed tooth profile rather than a cut-gear assumption.

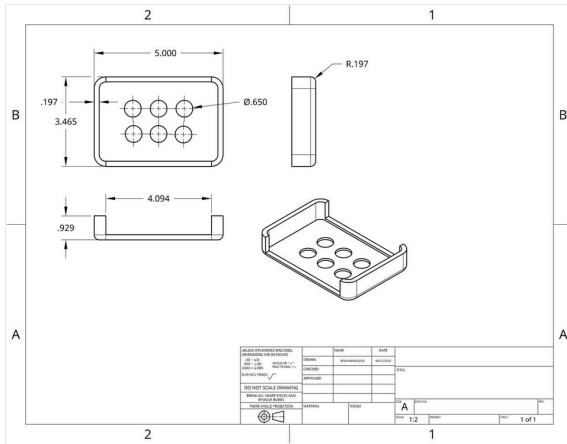


**Stepped rack.** Linear rack with two lead-in cam profiles at 1.000 in pitch (Scale 1:2), plus a keyed drive tab (.480/.680/.380 in, Scale 1:2) that engages the pin it lifts. This is the tolerated release version of the stepped-rack concept modeled earlier in Onshape.

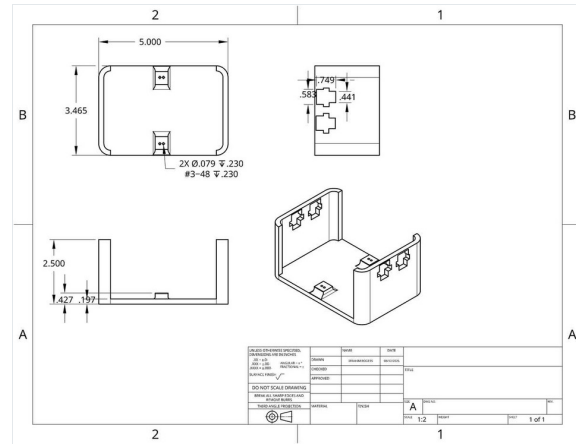


**Guide dowel.** Ø.100 in × 1.400 in ground pin. Locates the pin-guide block against the enclosure and keeps rack travel aligned under side load.

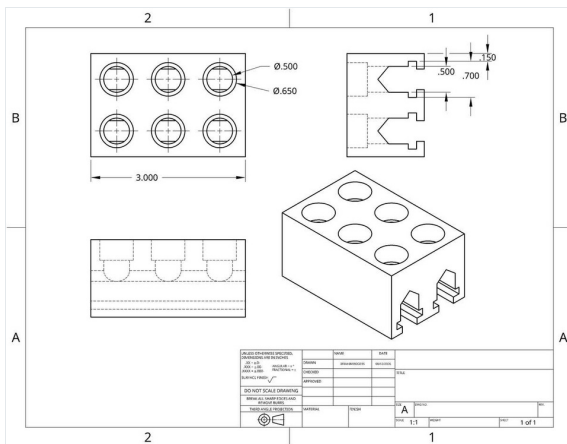
## ENCLOSURE & GUIDE STRUCTURE



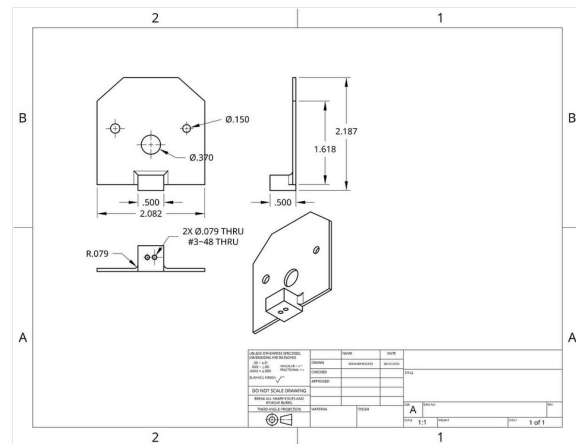
**Enclosure top cover.** 5.000 × 3.465 in with six  $\varnothing .650$  in access holes for the dome pins and R.197 in corner fillets. Mounting flange at .929/4.094 in locates the cover to the bottom chassis.



**Enclosure bottom chassis.** Matching 5.000 × 3.465 × 2.500 in shell with keyed mounting tabs (2×  $\varnothing .079$  in thru, #3-48 thru) for the pin-guide block and motor mount.



**Pin-guide block.** 3.000 in wide manifold with six bores ( $\varnothing .500$  in through,  $\varnothing .650$  in counterbore) carrying the dome pins, and snap-fit tabs (.150/.500/.700 in) that key it to the enclosure.



**Motor-side bracket.** L-bracket locating the drive motor relative to the pin-guide block, with  $\varnothing .370$  in motor-shaft clearance and two  $\varnothing .150$  in mounting holes on a .500 in tab.

These drawings belong to the enlarged six-pin build. Its clearance strategy does not carry over directly to the at-size cell, which uses printed features far finer than anything dimensioned here.

### KEY PROBLEM — TOLERANCE STACK-UP UNDER CYCLIC LOADING

Early prototypes jammed when actuated under side load. Failure tracking isolated two compounding causes:

- Guide-channel clearance too tight at 0.1 mm nominal, leaving insufficient margin once print tolerance and side deflection stacked.
- Dimensional creep in PLA after sustained cycling, drifting clearances further out of spec.

**Fix:** opened guide clearance to 0.15 mm, added lead-in chamfers, and moved to PETG. **Result:** jam rate fell from roughly 15% to under 2% across 2,000+ test cycles. The released drawing package on the preceding pages reflects this fix directly, including the  $\varnothing .500$  in pin bore sized to hold that clearance under load.

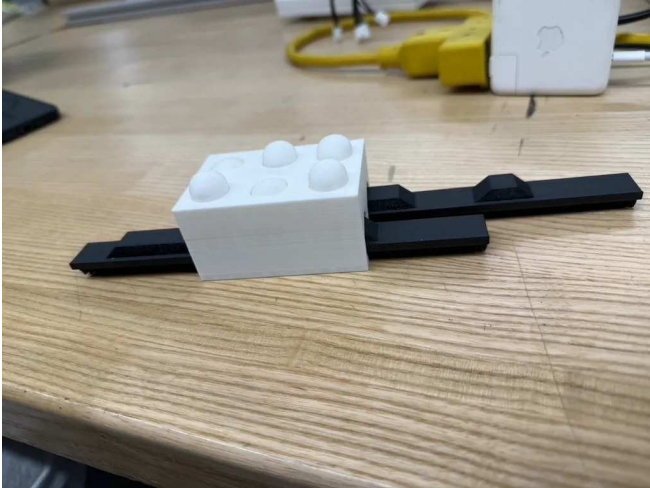
#### WHAT I TOOK FROM IT

- **Clearance is not just a manufacturing number.** It has to absorb print tolerance and accumulated wear, not only nominal fit.
- **Material choice is a lifecycle decision.** PLA was fine for form checks and wrong for cyclic loading, and that only showed up in testing.
- **Track failures, don't guess.** Logging jam events by condition pointed at side-load binding specifically, which made the fix targeted.
- **Design for assembly.** Consolidating cam and guide features cut part count from 12 to 8 and simplified the assembly sequence.
- **Cross-domain constraint.** Worked with the electrical side to keep actuation force inside servo torque limits while still meeting tactile force requirements.

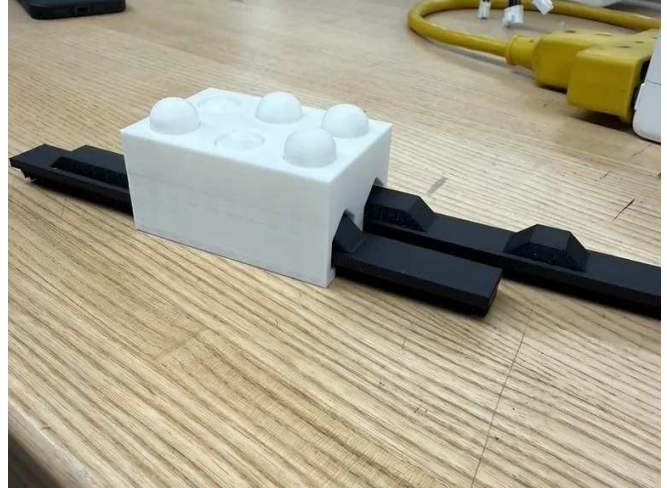


## PHYSICAL PROTOTYPE

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**Assembled prototype — side view.** White housing carries the pin actuation mechanism and rides on the black guide rails, which provide lateral constraint. Shows rail engagement and the travel available to the rack.



**Front three-quarter view.** Shows the six-dot face and the housing-to-rail interface where side-load binding originated. Lead-in chamfers on the rails were added at this interface to ease pin entry.

This PLA/PETG prototype predates the released drawing package on the preceding pages and is the build that surfaced the side-load binding described there.

ARCHITECTURE 3 — BEVEL GEARTRAIN & POSITIONAL ENCODING

Direct drive put the motor in line with the cell, which set a minimum pitch far wider than a reader expects and made tiling impossible. The final architecture moves the motors outboard and drives each cell through a bevel geartrain, then uses the two resulting racks to encode the whole character instead of lifting dots one at a time.

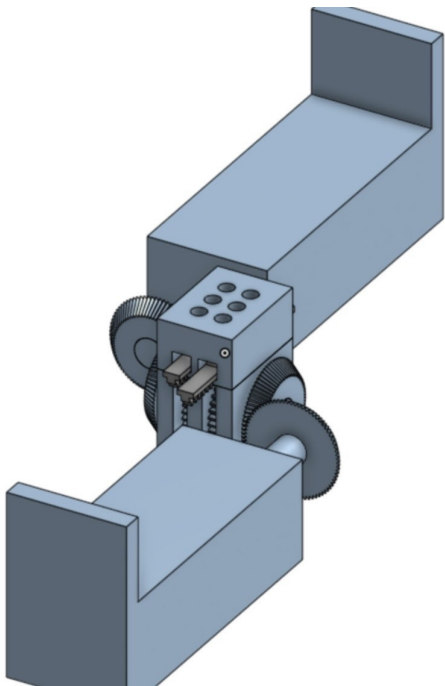
POSITIONAL ENCODING — 64 CHARACTERS, TWO MOTORS

The obvious architecture gives every dot its own actuator, six per cell. Instead, each cell carries two racks. Each rack is a stepped cam that encodes all eight raised/lowered combinations for its own set of three pins, and the motor drives the rack to whichever position the character calls for.

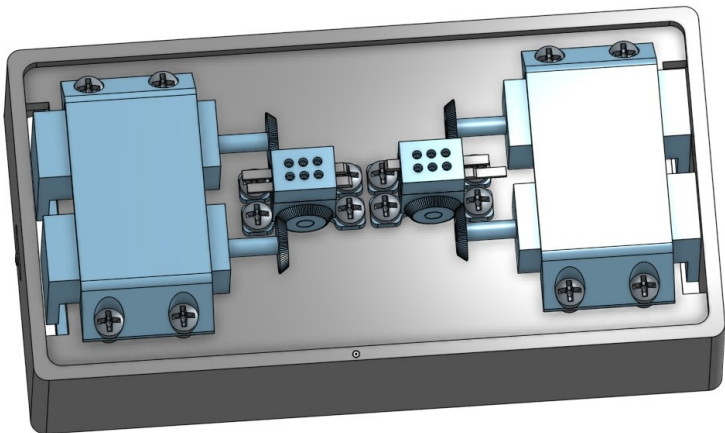
|               |   |   |   |   |    |    |    |     |
|---------------|---|---|---|---|----|----|----|-----|
| Rack position | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8   |
| Pins raised   | — | a | b | c | ab | ac | bc | abc |

Rack A carries dots 1–2–3, rack B carries dots 4–5–6. Eight positions against eight positions is 64 states, which is exactly the six-dot Braille set including the blank cell. Actuator count per cell drops from six to two.

CELL ARCHITECTURE & TILING



**Single cell.** The two racks enter from opposite faces, each driven by its own bevel gear. Rack channels extend away from the cell in opposite directions, which keeps the gearing and motors clear of the six-dot footprint.



**Two-cell assembly.** Cells sit adjacent at reading pitch with motors outboard and racks driving inward. Each cell is independently addressed, so a character can be rewritten without disturbing its neighbor.

Motors outboard and racks entering from opposite faces is what makes the cell tileable. Putting the actuator behind the dots would set a minimum cell pitch far wider than the 6.25 mm a reader expects, so the drive train had to leave the footprint entirely.



**Production-scale cell, printed parts, and drive motor.** The cell body sits on the output shaft of the gearmotor that drives it, with a rack, bevel gear, and dome pins alongside. Everything is shown at true relative scale, which is the point of the photograph: the drive hardware is larger than the cell it actuates.

### WHY THIS WAS THE HARD PART

Actuating six dots is not difficult on its own. Actuating them inside 6.25 mm, repeatedly, with hardware that can sit next to copies of itself, is the entire problem. The encoding scheme and the outboard drive layout are both answers to that constraint, not to the actuation.

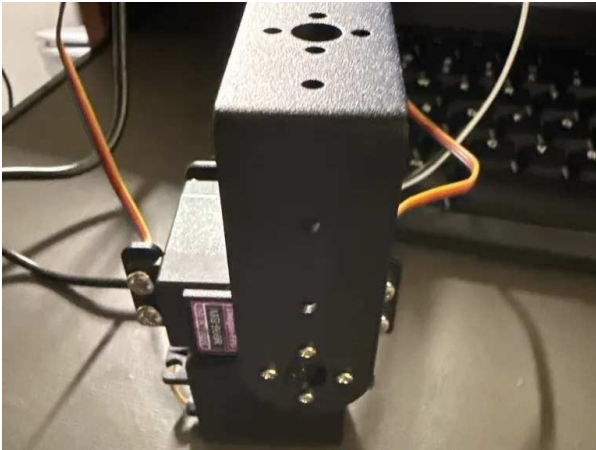
# Vision-Guided Pan/Tilt Tracking System

Independent Project · Dec 2025 – Present

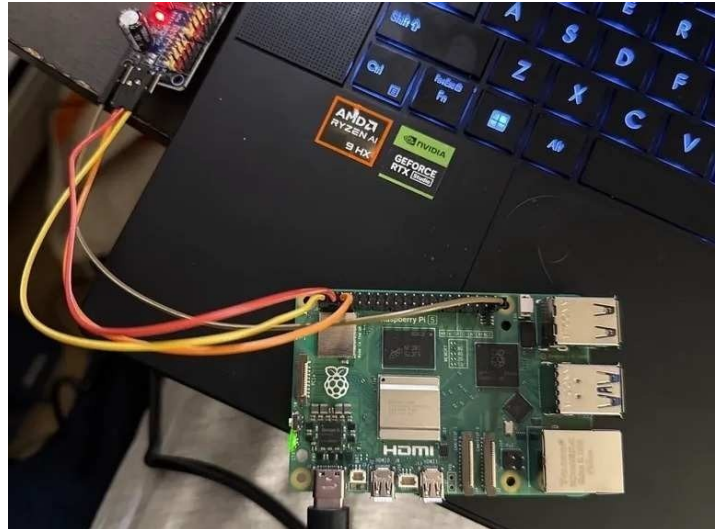
## OVERVIEW

**Goal:** Build a closed-loop target tracking platform — mechanical gimbal, electronics, and control software — that holds a marker centered in frame and recovers on its own when the target is lost.

**System:** Raspberry Pi 5, camera module, PCA9685 servo driver, dual-servo pan/tilt gimbal, OpenCV ArUco detection, Python control logic, Flask live video stream.



**Pan/tilt gimbal.** Custom two-axis 3D-printed housing with direct-drive servo coupling. No gear train, which removes backlash from the positioning path.



**Electronics integration.** Raspberry Pi 5 driving the PCA9685 servo controller over I<sup>2</sup>C, with servo power on a separate rail from compute power to keep actuation transients off the Pi.

## MECHANICAL DESIGN DECISIONS

- **Backlash:** direct-drive servo coupling with a press-fit interface, eliminating gear-train slop from the control loop.
- **Center of mass:** camera placement chosen to reduce servo torque load and improve response during fast target movement.
- **Rigidity:** gimbal frame reinforced to limit flex under rapid acceleration so the camera stays aligned during dynamic tracking.
- **Cable routing:** integrated to avoid wire interference across the full pan/tilt range.

## INTEGRATION DEBUG — CAMERA NOT ENUMERATING

### PROBLEM

No frame acquisition and no video feed. From the outside the whole system looked dead, with no indication whether the fault was mechanical, electrical, or software.

### PROCESS

- Used `raspistill` to separate camera hardware faults from software faults.
- Verified I<sup>2</sup>C device enumeration to confirm the Pi was seeing devices on the bus.
- Found improper CSI ribbon seating compounded by outdated camera firmware — two independent faults presenting as one symptom.
- Updated firmware and added error handling in Python for graceful frame-drop recovery.

### LESSON

Integration bugs concentrate at interface boundaries. Isolating with diagnostic tools beat guessing at the failure location, especially with two faults stacked.

## CONTROL IMPLEMENTATION

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- PID loop on pan and tilt axes to smooth servo motion and suppress oscillation around the target; gains tuned empirically on hardware.
- Automated re-acquisition via spiral search when the marker leaves frame or is occluded.
- Error handling for dropped frames, I<sup>2</sup>C communication failures, and servo limit violations.

### WHAT I TOOK FROM IT

- **Stage the integration.** Camera alone, then servos alone, then detection, then full loop. Bringing it up in stages saved hours over debugging everything at once.
- **Textbook PID isn't enough.** Real servo dynamics, friction, and loop latency meant the gains had to come from testing.
- **Cross-domain debugging is the actual skill.** Mechanical slop, power distribution, and software timing all produced similar-looking tracking errors.



# Gamma Shielding Test System

Co-Founder & COO — Melagen Labs, VC-backed aerospace materials startup · Jan 2024 – Sep 2024

## OVERVIEW

**Challenge:** Validate the shielding effectiveness of a melanin-HDPE composite against an aluminum reference using gamma radiation, with limited lab equipment and a startup budget.

**My Role:** Co-designed the test fixture, developed the experimental protocol for repeatable measurement, ran the testing, and performed the statistical analysis.

**Key Result:** Attenuation comparable to aluminum at roughly one-third the density, supported by ANOVA.

## TEST SYSTEM & SETUP



**Complete test system.** Fixed-geometry fixture holds a repeatable Cs-137 source-to-detector distance. A modular sample holder allows quick swaps between composite thicknesses without disturbing the measurement geometry.

### DESIGN REQUIREMENTS

- **Repeatability:** fixed source-to-detector positioning to minimize geometric variation between runs, the dominant error source in this measurement.
- **Modularity:** quick-change sample mounting to test multiple thicknesses and compositions efficiently.
- **Safety:** Cs-137 source containment and shielding per lab safety protocol.
- **Low-cost fabrication:** built from lab-available materials within startup budget.

## SAMPLE PREPARATION & MEASUREMENT



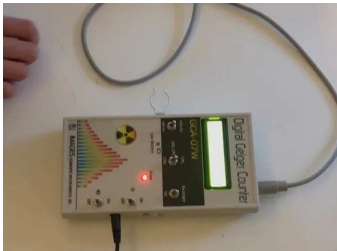
**Material preparation.** Melanin powder measured to a controlled mass so concentration stayed consistent batch to batch.



**Sample fabrication.** Composite cast in aluminum pans for consistent thickness, under controlled material-handling protocol.



**Test specimens.** Composite plates at varying thickness, measured and tested against aluminum references.



**Radiation measurement.** Calibrated Geiger counter at fixed distance from the shielded source, logging count rate for each shielding condition against an aluminum baseline.

## ENGINEERING PROCESS

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- **Fixture design for repeatability:** hand-positioning produced unacceptable run-to-run measurement scatter. Adding fixed mechanical alignment features cut that variation substantially and was the single biggest improvement to data quality.
- **Rapid iteration under constraints:** when the first fixture drifted after repeated sample changes, redesigned the mounting within days using available shop tools.
- **Statistical validation:** ANOVA across material types and thicknesses, with regression modeling of shielding effectiveness against melanin concentration.
- **Uncertainty analysis:** documented contributions from source activity calibration, detector efficiency, and geometric positioning so the results held up to outside review.
- **Technical communication:** translated the data into engineering presentations used in fundraising.

### WHAT I TOOK FROM IT

- **Fixture design drives data quality.** In experimental work, repeatability is the whole game. Small mechanical improvements to alignment had far more impact on confidence in the result than anything downstream in the analysis.
- **Constraints force decisions.** Limited budget and equipment meant choosing what actually mattered to measure well.

# Technical Skills

Tools and methods I can speak to in detail and have applied on the work in this portfolio

## CAD & DESIGN

Autodesk Inventor · SolidWorks · Onshape · AutoCAD  
GD&T (MMC/LMC) · worst-case tolerance stack-up · interference  
analysis · fixture design · engineering drawings · P&ID development

## PROGRAMMING & ANALYSIS

Python (OpenCV, machine-vision inspection, pandas) · C++ · MATLAB ·  
Excel  
ANOVA / regression · I<sup>2</sup>C and PWM interfaces · Raspberry Pi · Arduino ·  
hardware–software integration

## QUALITY & VALIDATION

Failure analysis · root-cause investigation · FMEA · dimensional  
inspection and metrology · experimental test design · nonconformance  
investigation · validation and test documentation

## FLUID & TEST SYSTEMS

Hydraulic/pneumatic architecture · pressure regulation and relief ·  
closed-loop recirculation · pump and chiller sizing · manifold design ·  
pressure-decay and flow-based leak test methods · test-stand  
architecture

## MANUFACTURING & PROTOTYPING

FDM 3D printing (PLA, PETG) · SLA/resin printing · rapid prototyping ·  
mechanical assembly · fixture fabrication · design for manufacturing ·  
design for assembly

## ENGINEERING COMMUNICATION

Failure-analysis reporting · design reviews · cross-functional  
requirements definition · vendor and supplier coordination · technical  
presentations to non-technical audiences

## ADDITIONAL PROJECT EXPERIENCE

### Assistive Sweeper Robot

Cornerstone II, London · Jan–Apr 2025

Co-designed an RC sweeper in SolidWorks,  
working drivetrain geometry against motor  
torque and ground clearance constraints.  
Iterated through a rapid prototyping sprint for  
operator usability and assembly.

### Autonomous Maze Robot

Cornerstone I, Oakland · Sep–Dec 2024

Programmed sensor-based autonomous  
navigation logic. Diagnosed drivetrain binding  
and electrical contact failures through  
systematic root-cause analysis across a two-  
week test cycle.

### Hydrogen Fuel Cell RC Vehicle

TKS Innovation Fellow · 2023–24

Designed a hydrogen fuel cell power system,  
managing power distribution and integration.  
Won a Google-sponsored pitch competition for  
the technical proposal.